

Machine Learning Group Project: Diabetes Risk Prediction

Group 3

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Introduction

- *Background*

Diabetes is a chronic condition that affects millions of individuals worldwide, with significant implications for both personal health and public healthcare systems. The prevalence of diabetes has increased substantially due to rising obesity rates, sedentary lifestyles, and other contributing risk factors. If left unmanaged, diabetes can lead to severe complications, including cardiovascular diseases, kidney failure, and nerve damage, placing a heavy burden on healthcare infrastructure.

- *Motivation*

Healthcare organizations and policymakers must develop effective strategies to manage and prevent diabetes. However, the complexity of risk factors (ranging from individual health conditions to lifestyle behaviors) makes this task challenging. Identifying key predictors of diabetes is essential to enable targeted interventions, optimize resource allocation, and improve patient outcomes. By leveraging data-driven insights, healthcare providers can implement evidence-based prevention programs to reduce the incidence of diabetes and improve overall public health.

- *Business Question*

How can lifestyle and behavioral changes be prioritized based on their impact on reducing diabetes risk across different demographic groups?

Data Description

- *Data Source*

The dataset used for this project is the Diabetes Health Indicators Dataset from Kaggle, specifically the `diabetes_binary_health_indicators_BRFSS2015`. It originates from the 2015 Behavioral Risk Factor Surveillance System (BRFSS) and consists of 253,680 records. Each record includes a binary diabetes outcome and multiple health indicators, such as BMI, physical activity levels, smoking habits, general health status, and demographic information (age, income, education levels, etc.).

- *Statistical Question*

What are the most significant predictors of diabetes, and how do they interact to influence disease likelihood?

This question allows us to prioritize lifestyle modifications for different demographic groups and create targeted prevention strategies.

- Utility of the Statistical Question

1. Helps determine which modifiable lifestyle factors (e.g., BMI, smoking, exercise) have the greatest impact on diabetes risk.
2. Provides a data-driven foundation for designing prevention strategies for specific demographic groups.
3. Identifies possible interactions between health indicators, improving understanding of how multiple risk factors contribute to diabetes.

- Limitations of the Statistical Question

1. The analysis is correlational, not causal—while we can identify strong predictors, we cannot definitively conclude that modifying a specific factor will prevent diabetes.
2. Certain external factors (e.g., genetics, environment, healthcare access) are not included in the dataset, potentially limiting the scope of the findings.
3. Some variables may be highly correlated, affecting model performance.

Exploratory Data Analysis

- *Overview*

- The dataset is highly imbalanced, with 85% non-diabetic and only 15% diabetic cases.
- No missing values were found, and all categorical variables were pre-encoded.
- 21 features were analysed.
- Feature correlation analysis (Figure A1, Appendix A) shows General Health (GenHlth) and Physical Health (PhysHlth) are strongly correlated (0.52).

2. Age and BMI Interaction on Diabetes Risk

Observations (Figure 4):

- Higher BMI values correlate strongly with an increased risk of diabetes, even for younger individuals. As BMI **exceeds 30 (obesity threshold)**, the probability of diabetes significantly rises.
 - Implications: Weight management should be a primary intervention, especially among younger obese individuals, to mitigate long-term diabetes risks.

Actionable Steps:

- Encourage regular health check-ups to monitor BMI trends.
- Implement targeted awareness campaigns focusing on healthy eating and exercise for high-BMI individuals.

3. Age and High Blood Pressure (HighBP) Interaction on Diabetes Risk

Observations (Figure 5):

- High blood pressure is a significant risk factor for diabetes, with its impact becoming more pronounced as individuals age.
- Individuals aged 40 and above with high blood pressure face a substantially increased risk, often exceeding **0.5** on the risk scale.
- Even among **individuals aged 60 and above**, those without high blood pressure maintain a **notably lower diabetes risk** compared to their hypertensive counterparts.
 - Implications: Effective hypertension management is essential to reducing diabetes risk, especially for individuals **aged 40 and above**. Preventive interventions should target this high-risk group.

Actionable Steps:

- Promote lifestyle modifications, including reduced sodium intake and increased physical activity.
- Implement routine hypertension screening, especially in middle-aged and elderly populations.

4. Age and High Cholesterol (HighChol) Interaction on Diabetes Risk

Observations (Figure 6):

- Individuals with high cholesterol (HighChol = 1) consistently exhibit higher diabetes risk across all age groups.
- While younger individuals (20-40) have a lower risk overall, those with high cholesterol still show an increased probability of developing diabetes.
- For those aged 40 and above, diabetes risk rises significantly, with high cholesterol further amplifying the risk in a continuous upward trend.
 - Implications: Cholesterol control through **dietary interventions and medication** can help mitigate diabetes risks.

Actionable Steps:

- Encourage routine lipid profile screenings.
- Advocate for heart-healthy diets, reducing saturated fat intake.

5. General Health and Difficulty Walking on Diabetes Risk

Observations (Figure 7):

- Poor general health (GenHlth = 1 or 2) is strongly associated with a higher diabetes risk, particularly for individuals with mobility difficulties (DiffWalk = 1), where the median risk approaches 0.5.
- Even among those with better health (GenHlth = 4 or 5), individuals with walking difficulties still face a higher risk compared to those without.
- Across all health categories, **those with mobility impairments consistently show elevated diabetes risk.**
 - Implications: Mobility limitations and overall poor health exacerbate diabetes risks, highlighting the need for targeted interventions.

Actionable Steps:

- Develop targeted physical therapy and rehabilitation programs for individuals with mobility challenges.
- Promote overall wellness through comprehensive health education.

Method & Results

- *Feature Pre-processing*

To prepare the dataset for modeling, we applied several preprocessing techniques:

- **Feature Scaling:** Continuous variables such as **BMI, Age, General Health, and Physical Health** were **standardized using *StandardScaler*** to ensure consistent feature contribution.
- **Class Imbalance Handling:** To address class imbalance, we used Logistic Regression with ***class_weight='balanced'***, which automatically adjusts the weight of each class to compensate for imbalance. This ensures that the model does not favor the majority class and improves predictive performance on the minority class.
- **Categorical Variable Processing:** Binary categorical features were already encoded, so no additional one-hot encoding was necessary.

- *Feature Selection:*

For feature selection, we used two key methods: **Feature Importance** from our Gradient Boosted Trees model and **SHAP Values** to understand each feature's impact on predictions. From these analyses, we identified the **top five most important features** affecting diabetes risk: **General Health (GenHlth), High Blood Pressure (HighBP), BMI, Age, and High Cholesterol (HighChol)**. The bar chart (Figure 2) shows Feature Importance, where HighBP and GenHlth have the strongest influence on the model. On the right, the SHAP Value plot (Figure 3) confirms this, showing how each feature contributes to predicting diabetes risk.

We also utilized **feature importance analysis from the Gradient Boosting Model**, selecting the **top 10 most influential features** based on their contribution to model predictions. This step ensured that only the most relevant variables were used, reducing noise and improving generalization (Figure 8).

- *Performance Metrics*

We evaluated the models using the following metrics:

- **Accuracy:** Measures overall correctness but can be misleading in imbalanced datasets.

- Precision: Ensures that predicted diabetes cases are correct (useful in minimizing false positives).
- Recall: Prioritized in healthcare applications to reduce false negatives, ensuring high-risk individuals are identified.
- F1-Score: Balances Precision and Recall, providing a comprehensive performance measure.
- ROC AUC Score: Measures the model's ability to distinguish between diabetic and non-diabetic individuals, making it a key metric in our evaluation.

Since healthcare applications prioritize **minimizing false negatives**, **Recall and ROC AUC were given more weight** when selecting the final model.

- *Hyperparameter Tuning*

Each model underwent hyperparameter tuning to enhance performance:

- Logistic Regression: Optimized regularization strength (C parameter) and adjusted class weights to balance sensitivity and specificity.
- Random Forest: Tuned number of trees (n_estimators), maximum depth, and min_samples_split to maximize predictive accuracy without overfitting.
- Gradient Boosting Model: Fine-tuned learning rate, number of boosting iterations (n_estimators), and max_depth to achieve the best trade-off between accuracy and generalization.

We used **GridSearchCV** and **RandomizedSearchCV** for hyperparameter optimization, ensuring robust performance across different parameter settings.

- *Model Selection*

Three supervised learning models were tested and compared:

1. Logistic Regression (ROC AUC = 0.81, see table 1):
 - Strengths: High interpretability, computational efficiency, and explainable coefficients.
 - Weaknesses: Linear assumptions limit its predictive power compared to non-linear models.

2. Balanced Random Forest (ROC AUC = 0.80, see table 2):
 - Strengths: Captures non-linearity, robust to overfitting, provides feature importance insights.
 - Weaknesses: Less interpretable than Logistic Regression, slower training time.
3. Gradient Boosting Model (ROC AUC = 0.82, see table 3):
 - Strengths: Highest accuracy, powerful for complex relationships, strong feature selection capabilities.
 - Weaknesses: Computationally expensive, harder to interpret.
4. Random Forest Model (ROC AUC = 0.81, see table 4):
 - Strengths: Handles complex patterns and interactions.
 - Weaknesses: Computationally expensive.
5. XGBoost Model (ROC AUC = 0.82, see table 5):
 - Strengths: Best balance of accuracy and efficiency.
 - Weaknesses: Requires significant computational power.
6. Decision Tree Model (ROC AUC = 0.80, see table 6):
 - Strengths: Highly interpretable.
 - Weaknesses: Prone to overfitting.

- *Final Model Selection & Deployment Considerations*

Since our dataset has fewer diabetes cases, we decided to use **class weighting** in the **Gradient Boosted Trees model**. This way, the model puts more importance on predicting diabetes cases correctly. After applying class weighting, we actually got the best ROC AUC score overall (Table 7), so we decided to go with Gradient Boosted Trees as our final model for the machine learning pipeline.

- I. **Superior Predictive Performance** – With the highest **ROC AUC score (0.826)** and the best balance between **precision (0.33)**, **recall (0.76)**, and **accuracy (0.73)**, it provides the most reliable predictions for diabetes risk.
- II. **Strong Generalization Ability** – The Gradient Boosting model captures complex relationships between risk factors while maintaining robustness against overfitting, making it suitable for real-world healthcare applications.

III. **Balanced Precision-Recall Trade-off** – While Logistic Regression offers greater interpretability, Gradient Boosting provides better overall classification performance, ensuring fewer missed high-risk cases while maintaining a reasonable false positive rate.

To further enhance predictive accuracy and fairness, future research could explore **feature selection optimization**, integrate **longitudinal health data** to capture risk progression over time, and experiment with **deep learning approaches** for nonlinear interactions. Additionally, **bias monitoring and fairness evaluation** will be essential to ensure ethical deployment, particularly when used for public health decision-making and clinical risk assessment.

Communication of Results and Advice to Public

Our analysis aimed to answer the question: How can lifestyle and behavioral changes be prioritized based on their impact on reducing diabetes risk across different demographic groups? By using machine learning models, we identified the most influential health and lifestyle factors that contribute to diabetes risk.

- *Key Findings & Prevention Recommendations*

Based on our final model, the most important factors associated with diabetes risk are:

1. BMI (Body Mass Index) – Higher BMI is one of the strongest predictors of diabetes. Maintaining a healthy weight through proper diet and exercise is essential for prevention.
2. General Health Perception – Individuals who rated their general health as poor were more likely to have diabetes. This suggests that early intervention in individuals experiencing multiple health issues could help reduce risk.
3. Age – Older individuals are more likely to develop diabetes, so routine screenings and preventive healthcare are especially important for people over 45.
4. Physical Health Issues – Those reporting frequent physical health problems had higher diabetes risk. Encouraging regular medical check-ups and chronic disease management can mitigate long-term complications.
5. Smoking – Current smokers had an increased likelihood of diabetes. Reducing tobacco consumption should be a key part of diabetes prevention programs.

Using these insights, health organizations, policymakers, and individuals can take proactive steps to prevent diabetes. Community-based initiatives, such as free exercise programs, public health campaigns on smoking cessation, and affordable health check-ups, should be prioritized to support at-risk populations.

- *Advice to the Public*

Taking small, sustainable steps can have a big impact on diabetes prevention. Here are some practical actions anyone can take today:

1. *Eat healthier*: Reduce processed foods, increase fiber intake, and choose whole grains over refined carbs.
2. *Monitor your health*: If you're over 45 or have a family history of diabetes, get regular check-ups and track key health indicators like blood sugar and BMI.
3. *Quit smoking*: Seek professional help or join a support group if needed.

By making these changes, individuals can take control of their health, reduce diabetes risk, and improve overall well-being. Prevention is always better than treatment, and small efforts today can lead to a healthier future.

- *Model Performance & Accuracy*

To ensure reliability, we tested multiple machine learning models, including Logistic Regression, Random Forest, and Gradient Boosted Trees. We first compared models based on ROC AUC to evaluate their overall performance in distinguishing between diabetic and non-diabetic individuals. While Gradient Boosting achieved the highest ROC AUC score, we selected Logistic Regression for its interpretability and reliability in a healthcare setting.

The model performed well in identifying high-risk individuals, with:

- **High ROC AUC score**, demonstrating strong overall classification performance.
- **High recall (sensitivity)** – Ensuring that most individuals at risk for diabetes were correctly identified, making it useful for early prevention.

Reference

Dataset:

Teboul, A. (2015). Diabetes Health Indicators Dataset
(diabetes_binary_health_indicators_BRFSS2015) [Data set]. Kaggle. Retrieved from
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https://www.cdc.gov/brfss/annual_data/2015/pdf/codebook15_llcp.pdf

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OpenAI. (2025). ChatGPT Generated Code and Model Improvements [AI-generated content]. Retrieved from
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<https://chatgpt.com/share/67a838f5-14f8-8000-84eb-e6040e89f1b6>

Appendix

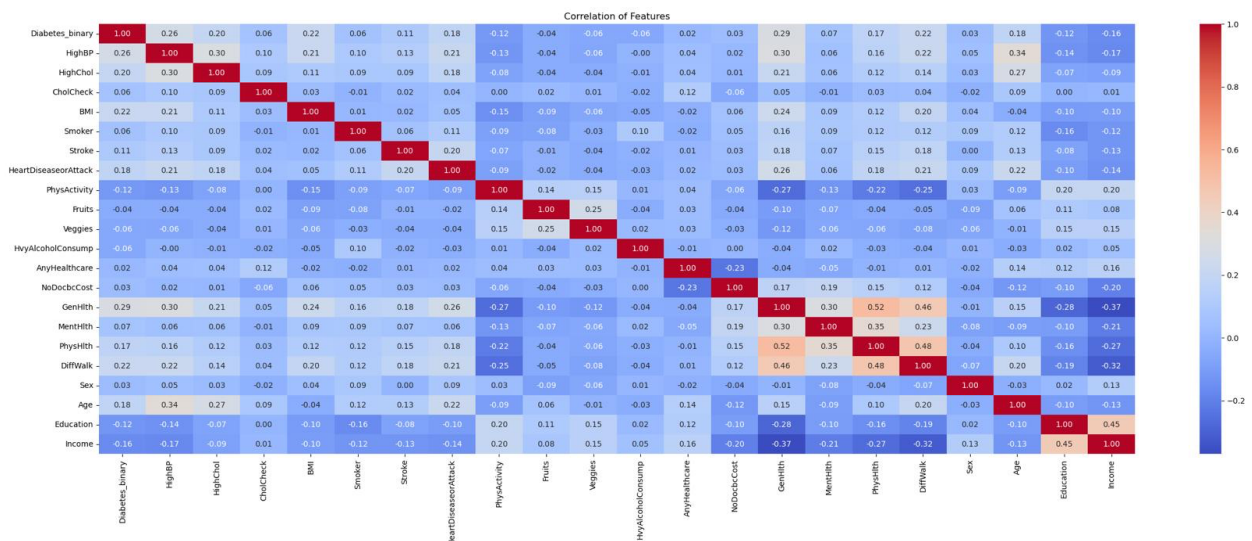


Figure 1: Correlation of Features

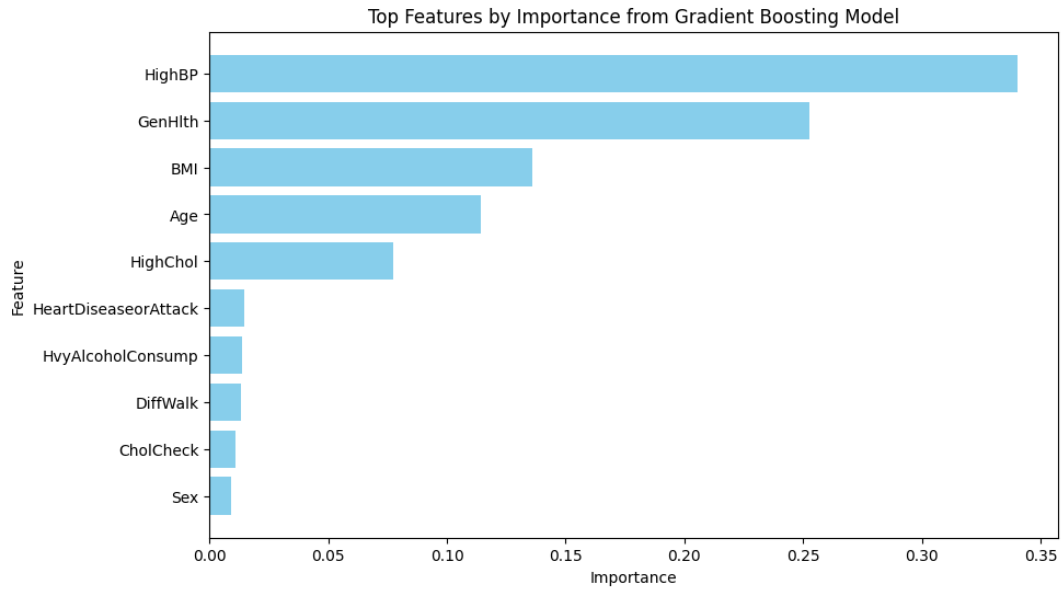


Figure 2: Top Feature by Importance from Gradient Boosting Model

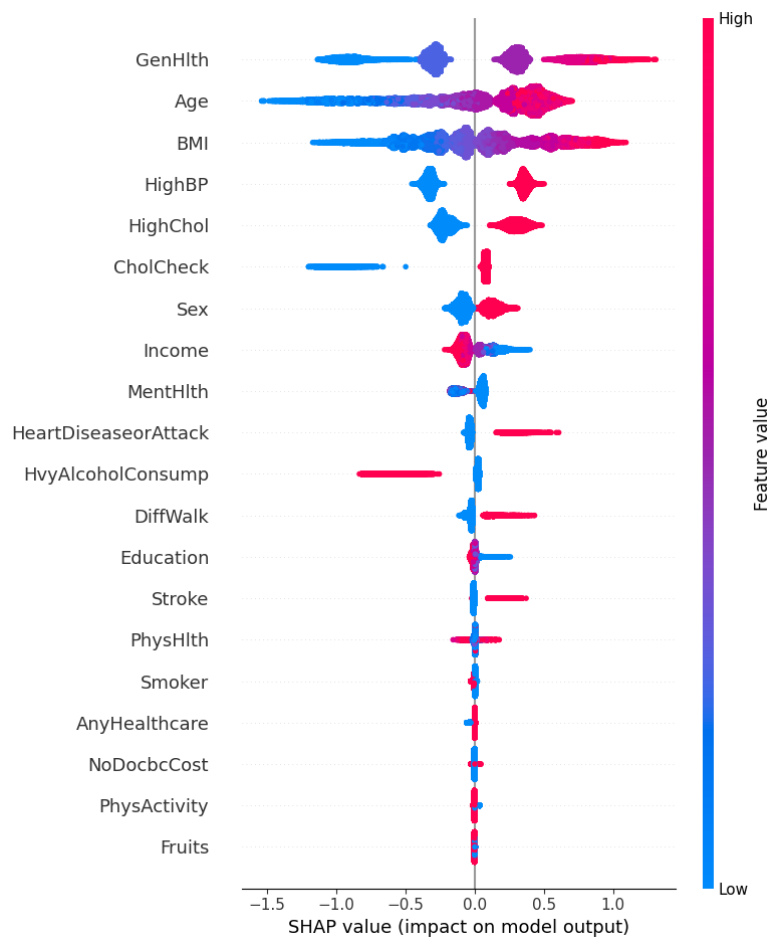


Figure 3: SHAP Value (Impact on model output)

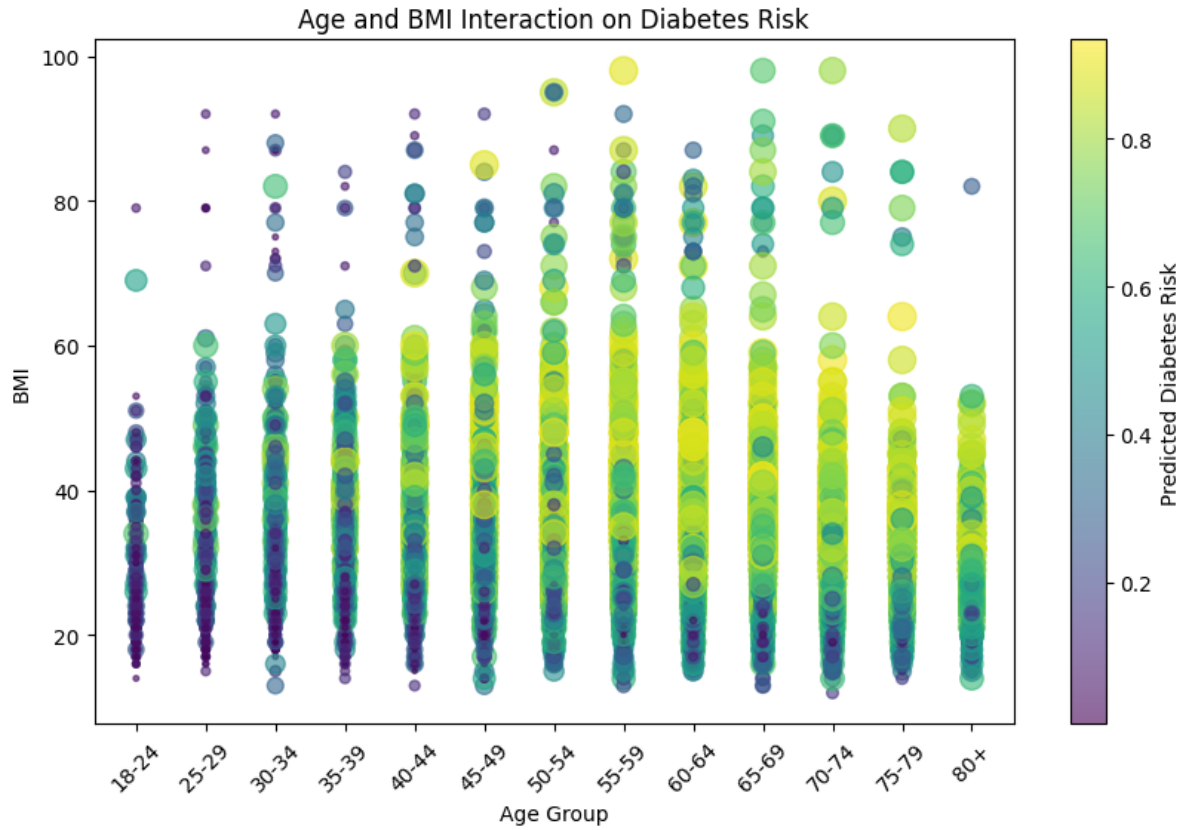


Figure 4: Age vs. BMI Interaction on Diabetes Risk

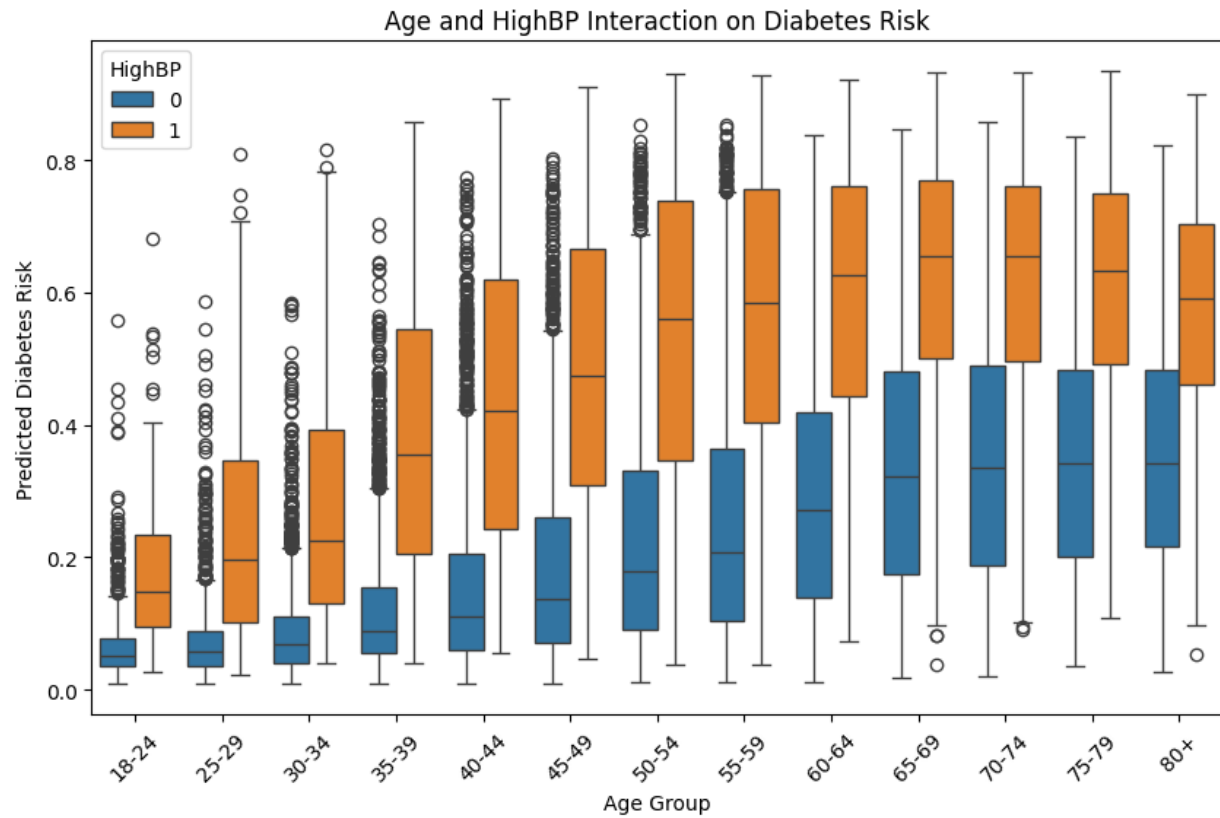


Figure 5: Age and HighBP Interaction on Diabetes Risk

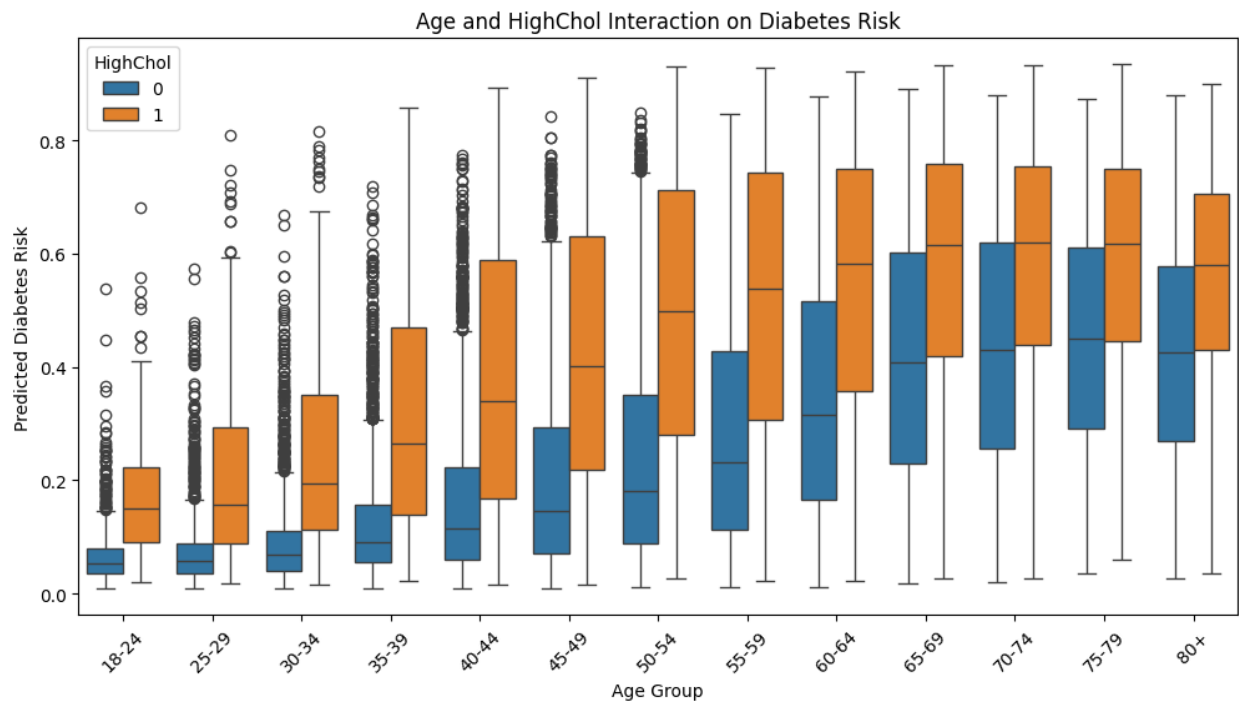


Figure 6: Age and HighChol Interaction on Diabetes Risk

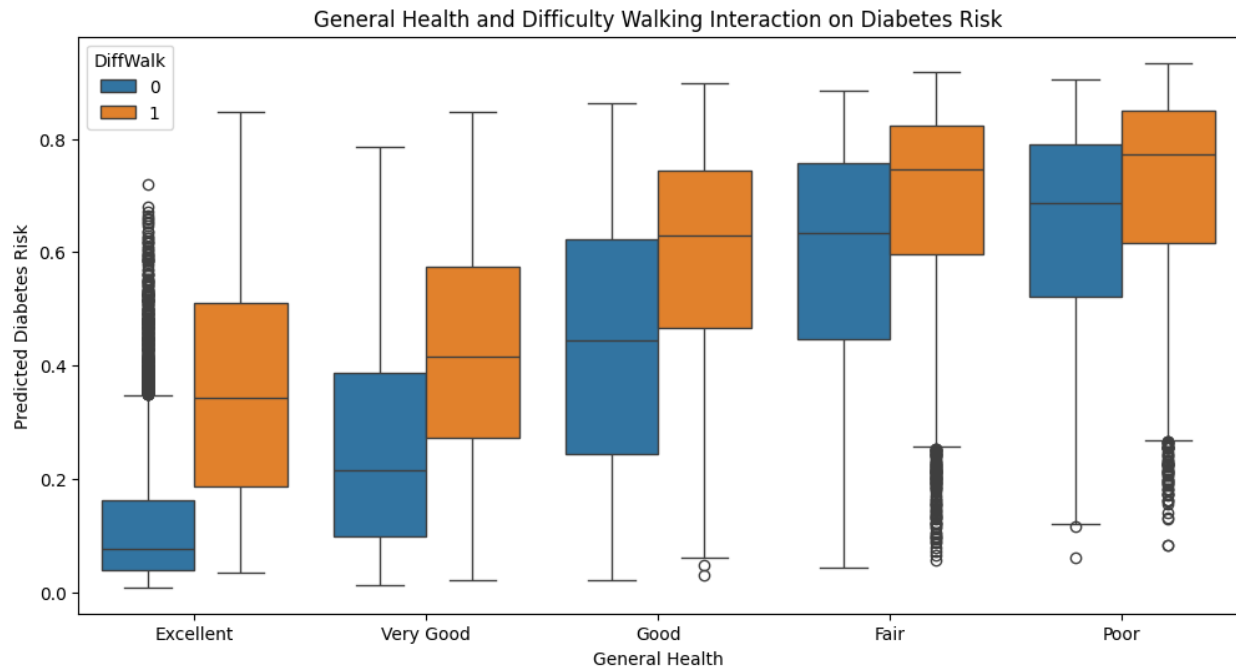


Figure 7: General Health and Difficult Walk Interaction on Diabetes Risk

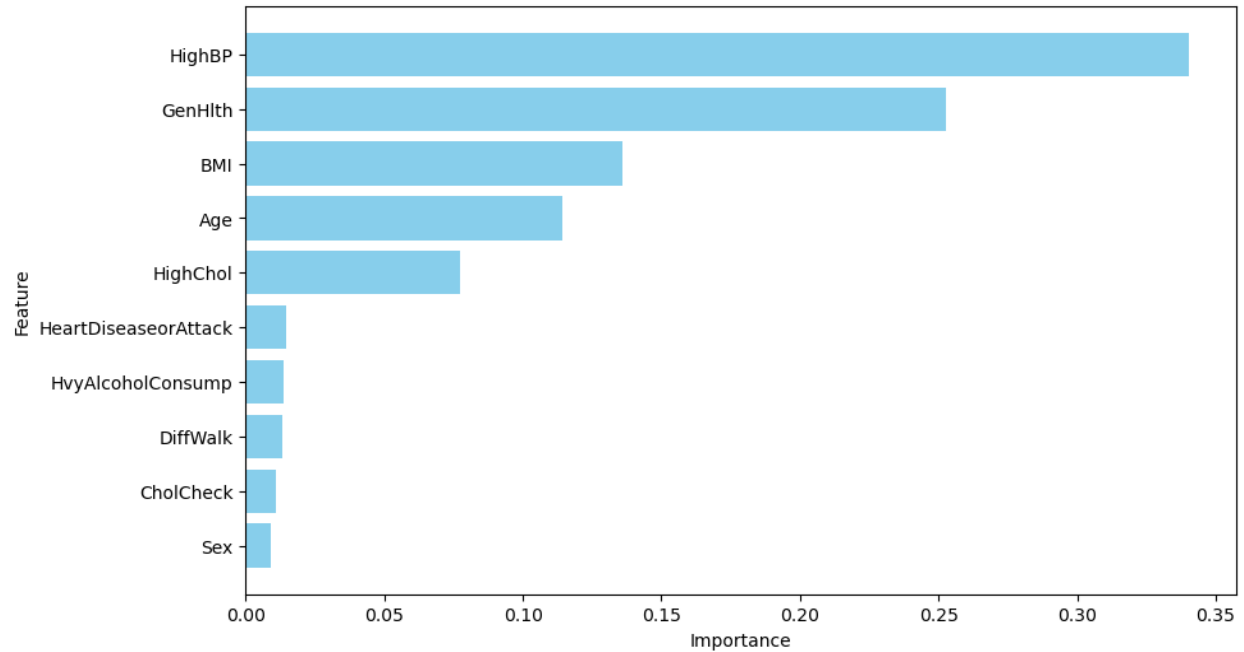


Figure 8: Top Features by Importance from Gradient Boosting Model

```

LogisticRegression classification report:
              precision    recall  f1-score   support

     0.0         0.95      0.73      0.82     43667
     1.0         0.31      0.76      0.44      7069

 accuracy          0.73     50736
 macro avg         0.63      0.74      0.63     50736
 weighted avg      0.86      0.73      0.77     50736

 ROC AUC: 0.8196422342353251
    
```

Table 1: Logistic Regression Model Performance Metrics

Balanced Random Forest classification report:				
	precision	recall	f1-score	support
0.0	0.95	0.71	0.81	43667
1.0	0.30	0.78	0.43	7069
accuracy			0.72	50736
macro avg	0.63	0.74	0.62	50736
weighted avg	0.86	0.72	0.76	50736
Balanced Random Forest ROC AUC: 0.818156407508059				

Table 2: Balanced Random Forest Model Performance Metrics

Gradient Boosted Trees classification report:				
	precision	recall	f1-score	support
0.0	0.88	0.98	0.93	43667
1.0	0.55	0.17	0.26	7069
accuracy			0.86	50736
macro avg	0.71	0.57	0.59	50736
weighted avg	0.83	0.86	0.83	50736
Gradient Boosted Trees ROC AUC: 0.8264158551273975				

Table 3: Gradient Boosted Trees Model Performance Metrics

Random Forest classification report:				
	precision	recall	f1-score	support
0.0	0.88	0.97	0.92	43667
1.0	0.49	0.18	0.26	7069
accuracy			0.86	50736
macro avg	0.68	0.57	0.59	50736
weighted avg	0.82	0.86	0.83	50736
Random Forest ROC AUC: 0.7969206130283785				

Table 4: Balanced Random Forest Model Performance Metrics

scale_pos_weight=3				
XGboost Classification Report:				
	precision	recall	f1-score	support
0.0	0.92	0.87	0.89	43667
1.0	0.40	0.55	0.46	7069
accuracy			0.82	50736
macro avg	0.66	0.71	0.68	50736
weighted avg	0.85	0.82	0.83	50736
XGboost ROC AUC: 0.8238746397615776				
scale_pos_weight=4				
XGboost Classification Report:				
	precision	recall	f1-score	support
0.0	0.94	0.81	0.87	43667
1.0	0.36	0.66	0.46	7069
accuracy			0.79	50736
macro avg	0.65	0.73	0.67	50736
weighted avg	0.86	0.79	0.81	50736
...				
macro avg	0.62	0.75	0.61	50736
weighted avg	0.87	0.69	0.74	50736
XGboost ROC AUC: 0.8241220043449048				

Table 5: XGBoost Model Performance Metrics

Decision Tree classification report:				
	precision	recall	f1-score	support
0.0	0.95	0.69	0.80	43667
1.0	0.28	0.77	0.42	7069
accuracy			0.70	50736
macro avg	0.62	0.73	0.61	50736
weighted avg	0.86	0.70	0.74	50736
Decision Tree ROC AUC: 0.8011051699632019				

Table 6: Decision Tree Model Performance Metrics

Gradient Boosted Trees classification report with weights:				
	precision	recall	f1-score	support
0.0	0.95	0.76	0.84	43667
1.0	0.33	0.73	0.46	7069
accuracy			0.75	50736
macro avg	0.64	0.75	0.65	50736
weighted avg	0.86	0.75	0.79	50736
Gradient Boosted Trees ROC AUC: 0.8264463298531641				

Table 7: Gradient Boosted Trees with Weights Performance Metrics